

Rev



- Learn & Remember
- Read only
- Doubt
- Special

Instruments Test Paper

1. A 2 axis gyro, measuring vertical changes will have:

- a one degree of freedom, vertical axis
- b two degrees of freedom, vertical axis
- c one degree of freedom, horizontal axis
- d two degrees of freedom, horizontal axis

2. The properties of a gyro are:

- 1. mass
- 2. rigidity
- 3. inertia
- 4. precession
- 5. rotational speed
- a. 1, 2, & 3
- b. 2 & 4
- c. 2 & 3
- d. 1 & 3

3. An aircraft fitted with a DRMC upon landing in a northerly direction will indicate:

- a no change
- b oscillation about north
- c a turn towards east
- d a turn towards west

RW 36 / RW 18

Eg.



4. Which of the following will effect a direct reading compass?

- 1. ferrous metals
- 2. non-ferrous metals
- 3. electrical equipment
- a. 1 only
- b. 1 & 3
- c. 1 & 2
- d. all 3

5. A vibrator may be fitted to an altimeter to overcome:

- a friction
- b hysteresis
- c lag
- d pressure error

6. An aircraft is flying at constant indicated altitude, over a warm airmass. The altimeter reading will be:

- a correct
- b greater than the real altitude
- c less than the real altitude
- d oscillating around the correct altitude

7. The mach meter consists of:

- a an airspeed indicator with mach scale
- b an airspeed indicator with an altimeter capsule
- c an altimeter corrected for density
- d a VSI and altimeter combined

8. CAS is IAS corrected for:

- a position and instrument error
- b instrument, pressure and density error
- c relative density only
- d compressibility

ICE
C
E
P
C
P

9. A DGI has:

- a one degree of freedom & a horizontal spin axis
- b two degrees of freedom & a vertical spin axis
- c two degrees of freedom & a horizontal spin axis
- d one degree of freedom & a vertical spin axis

10. An aircraft is flying at an indicated altitude of 16,000ft. The outside air temperature is -300 C. What is the true altitude of the aircraft?

- a 16,200 ft
- b 15,200 ft
- c 18,600 ft
- d 13,500 ft

ISA -17
TA = 16000 - 4.1/5000 x 13
ACTUAL = -30
DCU = -13
= 16200

11. The main cause of error in a DRMC is:

- a parallax in the rose
- b turning
- c magnetic deviation
- d latitude

12. QNH is:

- a the airfield barometric pressure
- b the setting that will give zero indication on the airfield
- c the equivalent sea level pressure at the airfield
- d the setting that will indicate airfield height

13. What is the Schuler period?

- a 21 minutes

- ☒ b 84 minutes
- ☐ c 1 oscillation in azimuth
- ☐ d 63 minutes

14. The vertical reference of a data generation unit is:

- ☐ a horizontal axis with 1 degree of freedom
- ☐ b vertical axis with 1 degree of freedom
- ☐ c horizontal axis with 2 degree of freedom
- ☒ d vertical axis with 2 degree of freedom

15. The torque motor of a gyro stabilised magnetic compass:

- ☒ a precesses the directional gyro
- ☐ b takes its input from the flux valve
- ☐ c moves the heading pointer
- ☐ d moves the Selsyn stator

16. A factor giving an error on a direct indicating compass would be:

- ☐ a crosswinds – particularly on east/west headings
- ☐ b parallax due to oscillations of the compass rose
- ☒ c acceleration on east/west headings
- ☐ d turning through east/west headings

17. A rate integrating gyro is used in:

- ☐ 1. inertial attitude unit
- ☐ 2. autopilot system
- ☐ 3. stabiliser servo mechanism system
- ☒ 4. inertial navigation unit
- ☐ 5. rate of turn indicator
- ☐ a 1, 2, 3, 4, & 5
- ☒ b 1 & 4
- ☐ c 2, 3, & 5
- ☐ d 2, 3, & 4

18. The errors of a DGI are:

- ☒ 1. earth rate
- ☒ 2. transport wander
- ☒ 3. banking when pitched up
- ☒ 4. annual movement of poles
- ☒ 5. mechanical problems
- ☐ a 2, 3, & 5
- ☐ b 3, 4, & 5
- ☒ c 1, 2, 3, & 5
- ☐ d all 5

19. An Air Data Computer (ADC) obtains altitude from:

- a. outside air temperature
- b. barometric data from static source
- c. time elapsed for signal to travel to and return from the earth
- d. difference between absolute and dynamic pressure

20. If the needle and the ball of a Turn & Slip indicator both show right, what does it indicate:

- a. turn to left & too much bank
- b. turn to right & too much bank
- c. turn to left & too little bank
- d. turn to right & too little bank

21. What formula gives the total temperature (TT) from the static temperature (TS):

- a $TT = TS (1 + 0.2 M^2)$
- b $TT = TS (1 + 0.2 K^2 M^2)$
- c $TT = TS / (1 + 0.2 K^2 M^2)$
- d $TT = TS (1 - 0.2 M^2)$

22. The Inertial Strapdown Unit of an IRS is programmed with co-ordinates during alignment in order to:

- a establish the tri-hedron with reference to the earth
- b establish true or magnetic heading
- c check the function of the laser gyros
- d compensate for aircraft movement

23. When descending through an isothermal layer at constant CAS, what does the TAS do?

- a increase at a linear rate
- b increase at an exponential rate
- c remain the same
- d decrease

24. What is VMO calculated from:

- a. CAS
- b. TAS
- c. COAS
- d. EAS

25. Descending from FL390 at maximum groundspeed, what will the pilot be limited by:

- a. VMO initially then MMO at a specified altitude
- b. MMO initially then VMO at a specified altitude
- c. VNE initially then MMO at a specified altitude
- d. VNO initially then VNE at a specified altitude

26. At constant weight, regardless of altitude, an aircraft always lifts off at a constant:

- a EAS
- b TAS
- c ground speed
- d CAS

27. VFE is the maximum speed that:

- a. the flaps can be operated
- b. the flaps may be extended in the take-off configuration
- c. the flaps may be extended in the landing configuration
- d. the flaps may be extended in a specified configuration

28. The white arc on the ASI indicates:

- a. VS1 at the lower end and VLE at the upper end
- b. VS0 at the lower end and VLE at the upper end
- c. VS0 at the lower end and VFE at the upper end
- d. VS1 at the lower end and VFE at the upper end

29. An ASI circuit consists of pressure sensors. The Pitot Probe measures:

- a total pressure & static pressure
- b dynamic pressure
- c static pressure
- d total pressure

30. Mach number is defined as the ratio of:

- a IAS to LSS
- b TAS to LSS
- c CAS to LSS
- d EAS to LSS

31. If a pitot source is blocked in an ASI, and the drain hole is blocked, but the static source is open, what will happen?

- a ASI reading goes to zero
- b ASI under reads
- c ASI over reads
- d ASI behaves like an altimeter

32. In a turn at constant angle of bank ... the rate of turn is:

- a. independent of weight and proportional a to TAS
- b. dependant on weight and inversely proportional to TAS
- c. independent of weight and inversely proportional a to TAS
- d. dependant on weight and proportional to TAS

TAS Increase
Rate of turn ↑

33. The Turn Indicator is a useful gyroscopic instrument. When used in association with an, attitude indicator will show:

1. angular velocity about the yaw axis
2. direction of turn
3. angular velocity about true vertical axis
4. speed of turn

a 1, & 3

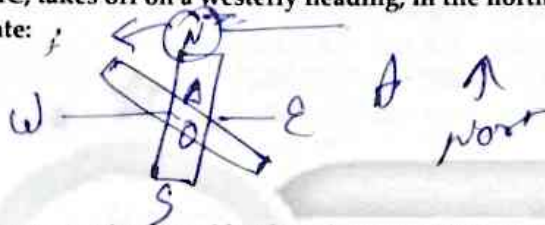
b 2, & 3

c 3, & 4

d 1, & 2

34. If an aircraft, fitted with a DRMC, takes off on a westerly heading, in the northern hemisphere, the DRMC will indicate:

- a a turn to the north.
- b oscillates about west
- c no turn
- d a turn to south



35. When turning through 90° at constant attitude and bank, a classic Artificial Horizon indicates:

- a nose up and correct angle of bank
- b attitude and bank angle are correct
- c nose up and bank angle too low
- d nose up and bank angle too high

36. The factors which will affect a Turn Indicator are:

1. angle of bank
2. aircraft speed
3. aircraft weight

a. all 3

b. 1 & 2

c. 1 & 3

d. 2 & 3

37. To obtain heading information from a Gyro Stabilised platform, the gyros should have:

- a 1 degree of freedom and a horizontal axis
- b 1 degree of freedom and a vertical axis
- c 2 degrees of freedom and a horizontal axis
- d 2 degrees of freedom and a vertical axis

38. What are the inputs to the ADC?

1. OAT
2. dynamic pressure

- 3. TAT
- 4. static pressure
- 5. electric power
- 6. pitot pressure
- 7. AOA
- a. 1, 2, 5 & 6
- b. all 7
- c. 3, 4 & 6
- d. 3, 4, 5, 6, & 7

39. The properties of a Turn Indicator are:

- 1. One degree of freedom
- 2. two degrees of freedom
- 3. two springs connected to the aircraft frame
- 4. spin axis in the longitudinal plane
- 5. spin axis parallel to the yaw axis
- 6. spin axis horizontal
- a. 1, & 6
- b. 2, & 5
- c. 1, & 4
- d. 2, & 6

40. A gravity erector system corrects errors on a:

- a. DGI
- b. artificial horizon
- c. turn indicator
- d. RIMC

41. In a Gyro magnetic Compass the flux gate transmits information to the:

- a heading indicator
- b amplifier
- c error detector
- d erecting system

42. VNO is the max. speed which:

- a the pilot can fully deflect the controls.
- b should only be exceeded in still air and with caution.
- c should never be exceeded.
- d must not be exceeded for flap/gear extension

43. If while level at FL 270, at a constant CAS, temperature falls, what happens to the Mach

- No.
- a decreases.

- b increases.
- c remains constant.
- d increases depending on whether temp > ISA or < ISA.

44. If the static vent becomes blocked on an unpressurised a/c, what could you do?

- a open the window.
- b break the VSI glass.
- c compute altitude mathematically.
- d Select standby pitot source

45. What does the "barbers pole" on an ASI indicate?

- a VMO & altitude.
- b VMO & temperature.
- c VNO
- d VNE

46. On board a/c, true altitude shown from:

- a standard atmosphere.
- b pressure altitude.
- c density altitude.
- d temperature altitude.

47. On a turn and slip indicator, needle to the left and ball to the right indicates:

- a turn to the right, not enough bank.
- b turn to the left, too much bank.
- c turn to the left, not enough bank.
- d turn to the right, too much bank.

48. What is density altitude:

- a altitude in the standard atmosphere at which the prevailing density is equal to the density in the standard atmosphere
- b pressure altitude corrected for prevailing temp.
- c temperature altitude.
- d pressure corrected

49. A radio altimeter is:

- a ground based and measures true altitude.
- b ground based and measures true height.
- c a/c based and measures true altitude.
- d a/c based and measures true height.

50. An a/c is travelling at 120 kts, what angle of bank would be required for a rate 1(one) turn:

- a. 300

$$\frac{120}{19} + 7 = 19$$

- b. 120
- c. 180
- d. 350

51. An a/c is travelling at 100 kts forward speed on a 30 glideslope. What is its rate of descent?

- a 500 ft/min.
- b 300 ft/min.
- c 250 ft/min.
- d 600 ft/min.

$$\text{Rate} = \frac{\text{gs slope angle} \times \text{forward speed}}{60}$$

52. If the pitot tube is leaking (and the pitot drain is blocked) in a non-pressurised a/c, the ASI will:

- a under-read.
- b over-read.
- c over-read in the climb, under-read in the descent.
- d under-read in the climb, over-read in the descent.

53. An RMI rose is mechanically stuck on 090 degrees. The ADF pointer indicates 225 degrees. What is the relative bearing to the beacon?

- a 225 degrees.
- b 135 degrees.
- c Cannot be determined.
- d 000 degrees.

$$\text{Radial } 225 - 90 = 135$$

R.B. Unknown

54. Using a classic Artificial Horizon, the a/c performs a right turn through 270 degrees at a constant angle of bank and rate of turn. The indication is:

- a Nose up, too much bank.
- b Nose up, not enough bank.
- c Nose up, wings level.
- d Bank and pitch correct.

55. In a DGI what error is caused by the gyro movement relative to the earth?

- a. Earth Rate
- b. Transport Wander
- c. real wander
- d. latitude error

56. In a right turn while taxiing, the correct indications are:

- a Needle left, ball right.
- b Needle left, ball left.
- c Needle right, ball right.
- d Needle right, ball left.

57. An aircraft is taking off on a runway heading 0450, in still air, with a compass having 00 deviation. The runway is on an agonic line. What are the northerly turning errors (NH)?

- a. compass moves to less than 0450
- b. compass moves to more than 0450
- c. compass stays on 0450 if wings are kept level
- d. compass remains on 0450

58. True heading can be converted into magnetic heading using a compass and:

- a A map with isogon lines. *same magnetic declination*
- b A map with isoclinical lines.
- c A deviation card.
- d A deviation curve

59. At sea level ISA, TAS:

- a Equals CAS
- b Is greater than CAS
- c Is less than CAS

60. What will the altimeter read if the layers beneath the aircraft are all colder than standard?

- a read lower than the real altitude
- b read higher than the real altitude
- c read the correct altitude
- d readings will fluctuate

61. The flux valve in a RIMC

- a is supplied with AC current (usually 487.5 Hz).
- b is fed with DC.
- c is made of perm-alloy magnetic steel.
- d has its own self exciter unit.

62. The indications of a machmeter are independent of:

- a Temperature (OAT)
- b Static Pressure
- c Differential static and dynamic Pressure
- d Dynamic Pressure

63. An artificial horizon has:

- a 1 degree of freedom and an horizontal axis.
- b 2 degree of freedom and an horizontal axis.
- c 1 degree of freedom and a vertical axis.
- d 2 degree of freedom and a vertical axis.

64. The rigidity of a gyro is improved by:

- a Increasing RPM and concentrating the mass on the periphery of the rotor.
- b Increasing RPM and concentrating the mass at the hub of the rotor.
- c Decreasing RPM and concentrating the mass on the periphery of the rotor.
- d Decreasing RPM and concentrating the mass at the hub of the rotor.

65. What is the speed of sound at sea level ISA

- a 644kts.
- b 661kts.
- c 1059 kts
- d 583kts.

66. What is the speed of sound at 25,000 ft and -28°C .

- a 624kts.
- b 618kts.
- c 601kts
- d 610kts.

235 - 3957k

-28 + 273

67. What is the speed of sound at 30,000 ft and -40°C .

- a 562kts.
- b 595kts.
- c 590kts.
- d 661kts.

68. If a constant CAS is maintained in a climb, what happens to the mach number

- a. remains constant
- b. increases
- c. decreases

69. A compass swing is used to:

- a align compass north with magnetic north.
- b align compass north with true north.
- c align magnetic north with true north.
- d get true north and lubber line aligned.

70. The TAT probe measures TAT by:

- a $\text{TAT} = \text{SAT} + \text{kinetic heating}$.
- b $\text{TAT} = \text{SAT} - \text{heating due to compressibility}$.
- c $\text{TAT} = \text{SAT} - \text{kinetic heating}$.
- d $\text{TAT} = \text{SAT} + \text{heating due to compressibility}$.

71. If a pitot tube and drains are blocked at altitude by icing, during a descent the ASI will:

- a read constant airspeed.
- b under read.

- c over read.
- d show zero.

72. An IRS is aligned when turned on so as to:

- a calculate the computed trihedron.
- b establish true and magnetic north.
- c establish position relative to true north and magnetic north.
- d establish magnetic north.

73. The advantages of an ADC over a traditional pitot - static system (list)

- 1. Position and compressibility correction.
- 2. reduced lag
- 3. ability to supply many instruments
- 4. ability to act as an altimeter following failure.
- 5. (apparently all the answer bar one had 4 as a correct answer)
- a. 1, 2 & 3
- b. 1, 2 & 4
- c. 2, 3 & 4
- d. 1, 3 & 4

74. The frequency band used for a Radio Altimeter is:

- a. SHF
- b. VHF
- c. UHF
- d. LF

75. What is the purpose of the latitude nut in a DGI?

- a. to correct for latitude error
- b. to correct for transport wander
- c. to correct for earth rate
- d. to correct for coriolis error

76. Total Air Temp is always ____ than Static Air Temp and the difference varies with ____.

- a warmer, altitude.
- b colder, altitude.
- c warmer, CAS.
- d colder, CAS.

77. In a slightly banked turn, the turn needle will indicate:

- a roll rate.
- b rate of yaw.
- c angular velocity about the vertical axis.
- d rate of pitch.

78. The Primary Flying Display (PFD) displays information dedicated to:

- a. engine data and alarms
- b. flight path
- c. weather radar
- d. aircraft systems

79. What are the inputs to the FMS?

- 1. Radio Aids
- 2. Engine Parameters
- 3. Air Data
- 4. Route Data
- 5. Terminal Data
- 6. Operating Data
- a. 1, 3, 4 & 6
- b. 2, 3, 4, & 5
- c. All of the above
- d. 1, 2, 3 & 6

80. What are the upper and lower limits of the yellow arc on an ASI?

- a. lower limit VLO and upper limit VNE
- b. lower limit VLE and upper limit VNE
- c. lower limit VNO and upper limit VNE
- d. lower limit VLO and upper limit VLE

81. What does the blue line on an ASI of a twin propeller engine aircraft indicate?

- a. VYSE
- b. VNO
- c. VFE
- d. VMCA

82. The gravity erecting device on a vertical gyro is used on which instrument;

- a. directional gyro unit
- b. turn indicator
- c. artificial horizon
- d. gyromagnetic device

83. In a VSI lag error is improved by:

- a. bi-metallic strip
- b. two
- c. use of an accelerometer system
- d. return spring

84. An aircraft fitted with a DRMC is landing in a southerly direction, in the Southern Hemisphere. What indications will be seen on the DRMC?

- a. 1800 turn to east
- ☒ b. no apparent turn
- c. turn to west

85. What is the maximum drift of a gyro, due to earth rate:

- a. 900 per hour
- b. 1800 per hour
- ☒ c. 150 per hour
- d. 50 per hour

86. An aircraft is flying a true track of 3600 from 50 south to 50 north. What is the change in apparent wander rate:

- a. 00 per hour
- ☒ b. + 50 per hour
- c. - 50 per hour
- d. depends upon groundspeed

87. When turning through 1800 at constant attitude and bank, a classic Artificial Horizon indicates:

- ☒ a. nose up and correct angle of bank
- b. attitude and bank angle are correct
- c. nose up and bank angle too low
- d. nose up and bank angle too high

88. What is the Schuler period?

- a. 48 minutes
- b. 84 seconds
- ☒ c. 48 seconds
- d. 84 minutes

89. Mach number is defined as:

- a. The ratio of pitot pressure to dynamic pressure
- b. The ratio of static pressure to dynamic pressure
- ☒ c. The ratio of dynamic pressure to static pressure
- d. The ratio of static pressure to pitot pressure

90. You are flying at a constant FL 290 and constant mach number. The total temperature increases by 50. The CAS will:

- ☒ a. remain approximately constant
- b. increase by 10 kts
- c. decrease by 10 kts

Because Mach and CAS doesn't depend on temp.

d. will increase or decrease depending on whether you are above or below ISA.

91. An aircraft turns from south-west to south-east when situated at 450N, what heading should you roll out on if using a DRMC?

- a. 1300
- b. 1150
- c. 1400
- d. 1550

92. What is SAT?

- a. relative temperature measured in K
- b. differential temperature measured in K
- c. relative temperature measured in 0C
- d. ambient temperature measured in 0C

93. If an aircraft climbs, at constant mach No, in ISA conditions what happens to the TAS and the CAS?

- a. TAS increases and CAS increases
- b. TAS remains constant and CAS decreases
- c. TAS decreases and CAS increases
- d. TAS decreases and CAS decreases

94. Where is the earth rate wander, and the transport wander of a gyro equal to zero?

- a. North Pole
- b. Equator
- c. 450 N
- d. 450 S

95. What happens when the static pressure supply, to an altimeter, becomes blocked during a descent?

- a. reduces to zero
- b. over reads
- c. under reads
- d. indicates altitude at which blockage occurred

96. What happens when the static vent supplying an ASI is blocked, and the ram air inlet remains clear?

- a. ASI acts opposite to an altimeter
- b. ASI always over reads / reads a higher value
- c. ASI always under reads / reads a lower value
- d. ASI acts like an altimeter

97. In a left turn while taxiing, the correct indications are:

- a Needle left, ball right.
- b Needle left, ball left.
- c Needle right, ball right.
- d. Needle right, ball left.

98. VLO is defined as:

- a. the maximum speed at which to fly with the landing gear retracted
- b. the maximum speed at which the landing gear may be retracted or extended
- c. the maximum speed at which to fly with the landing gear extended
- d. the minimum speed at which to fly with the landing gear extended

99. VNE is defined as:

- a. the speed which must not be exceeded in still air, or without caution
- b. the speed above which the landing gear may not be extended
- c. the speed which must never be exceeded
- d. the maximum speed for normal flap extension to be selected

100. In a left turn, the ball of the turn co-ordinator is out to the right, what corrective action is required?

- a. more right rudder
- b. less right bank
- c. more left bank
- d. more left rudder

101. In a gyro magnetic compass, where does the torque motor get its information from?

- a. the flux gate
- b. error detector
- c. the rotor gimbal
- d. amplifier

102. If an aircraft is descending at constant mach number, and the total air temperature remains constant, what happens to the CAS?

- a. remains constant
- b. decreases
- c. increases
- d. increases if the temperature is below standard, and decreases if the temperature is above standard

103. What are the advantages of a laser gyro compared to a conventional gyro?

- a. has a longer cycle life
- b. takes longer to set up/ spin up
- c. uses more power
- d. takes longer to align

104. A machmeter measures the ratio of;

- a. pitot pressure to static pressure
- b. (pitot pressure minus static pressure) to static pressure
- c. pitot pressure times static pressure
- d. pitot pressure to (static pressure times pitot pressure)

105. Which instrument has a 2° rotation in the horizontal axis?

- a. artificial horizon
- b. flux detector
- c. directional gyro indicator ✓
- d. turn indicator

106. The maximum drift error sensed by an uncompensated DGI will be:

- a. 15° per hour
- b. 30° per hour
- c. 45° per hour
- d. 60° per hour

107. The green arc on the ASI is used to identify which speed range:

- a. V_{SO} to V_{NO}
- b. V_{S1} to V_{FE}
- c. V_{S1} to V_{NO}
- d. V_{S1} to V_{LO}

108. Pressure altitude may be defined as:

- a. lowest forecast regional pressure
- b. pressure measured in the standard atmosphere
- c. altitude indicated with QFE set on the altimeter
- d. altitude indicated with QNH set on the altimeter

109. What is the effect on an altimeter reading if variations in static pressure occur near to the pressure source?

- a. a change in hysteresis error
- b. a change in the instrument error
- c. a change in the position error
- d. a change in the compressibility error

110. What is the value of the angle of magnetic dip at the South Pole?

- a. 0°
- b. 45°
- c. 90° ✓
- d. 60°

turning error other change in 35°

111. A standby artificial horizon must have the following properties:

- 1. a remote gyro
- 2. its own power supply
- 3. only to be used in emergency
- 4. its own gyro
- 5. one for each certified pilot
- a. all the above
- b. 1, 3, & 5
- c. 2, 3, & 4
- d. 2 & 4

112. During a descent at constant CAS and total temperature, the mach no:

- a. increases
- b. remains constant
- c. increases if SAT is greater than standard temperature and decreases if it is lower
- d. decreases

113. The single most significant item which makes a servo altimeter more accurate is:

- a. electromagnetic pick-off
- b. logarithmic scale
- c. temperature compensated spring
- d. multiple pointers

114. Which of the following gyro instruments has one degree of freedom?

- a. artificial horizon
- b. turn indicator
- c. directional gyro
- d. slaved gyro compass

115. If a large aircraft is slide slipped to starboard, and the port static vent is blocked, what will the altimeter read?

- a. under read
- b. read correctly
- c. Over read
- d. fluctuate

116. Mach number is determined from: (PT = total pressure, PS = static pressure)

- a. $(PT + PS) \times PT$
- b. $(PT - PS) \times PS$
- c. $(PT \times PS) \times PT$
- d. $(PT - PS) / PS$

117. The right static vent is blocked, when the aircraft yaws to the right. Does the altimeter:

- a. Over read
- b. under read
- c. unaffected

118. If the radio altimeter fails:

- a. height information disappears
- b. aural warning given
- c. radio alt flag, red lamp, and aural warning given
- d. radio alt flag and red lamp activates.

119. During a descent at a constant mach number, there is an increase of total temperature by 50. What effect does this have on CAS?

- a. remains almost constant
- b. increases if SAT is more than standard and decreases if SAT is less than standard
- c. increases by 10 kts
- d. decreases by 10 kts

120. VNO is defined as:

- a. maximum structural cruising speed
- b. never exceed speed
- c. manoeuvring speed
- d. maximum operating speed

121. If the left static vent is blocked, and the right static vent is clear. What will the altimeter read if the aircraft maintains constant level?

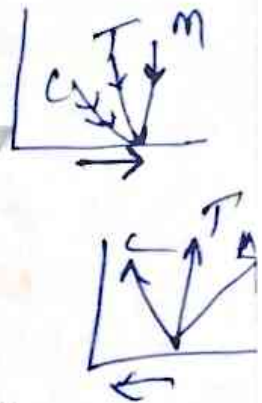
- a. read correctly whatever the situation
- b. under read
- c. if side slipping to the left, altimeter will over read.
- d. if side slipping to the right, altimeter will over read.

122. An aircraft is flying at constant indicated altitude, over a cold airmass. The altimeter reading will be:

- a. greater than the real altitude
- b. standard altitude
- c. same as the real altitude
- d. less than the real altitude

123. The machmeter compares: (PT = total pressure, PS = static pressure)

- a. (PT - PS) to PT
- b. (PT + PS) to PS
- c. (PT - PS) to PS
- d. PT to PS



124. From where does the air data computer (ADC) obtain aircraft altitude?

- a. OAT probe
- b. dynamic – absolute ambient pressure
- c. absolute barometric sensor on aircraft fuselage
- d. IRS

125. An aircraft is accelerating to take-off on a runway with a QDM of 045° . Which way does the DRMC move, if the aircraft is in the Northern Hemisphere?

- a. less than 45°
- b. more than 45°
- c. correct if wings are level
- d. correct

126. When turning right onto north, through 900 , what heading on your DIC should you roll out on, if the aircraft is in the Northern Hemisphere?

- a. 020°
- b. 360°
- c. 340°
- d. 320°

127. What does a radio altimeter, for an aircraft in the landing configuration, measure:

- a. height of aircraft wheels above the ground
- b. height of the aircraft above the ground
- c. altitude of the aircraft
- d. altitude of the aircraft wheels

128. Why is a servo altimeter better than a sensitive altimeter?

- a. it has a pick-off coil
- b. it is more accurate at low level
- c. it has ambient pressure in the capsule
- d. it is fitted with a knocking device

129. In an altimeter what is fed to: the capsule (i) and to the case (ii)?

- a. (i) vacuum (ii) static input
- b. (i) static input (ii) vacuum
- c. (i) pitot input (ii) static input
- d. (i) total input (ii) ambient input

130. What principle does the radio altimeter work on?

- a. pulse modulation
- b. amplitude modulation
- c. pulse modulation and carrier wave
- d. frequency modulation and carrier wave

131. What is indicated on the ASI when the static vent blocks during a descent?

- a. under reads
- b. reads correctly
- c. over reads
- d. reads zero

132. A rate integrating gyro is used in:

- a. inertial attitude unit
- b. autopilot system
- c. inertial navigation system
- d. a rate of turn indicator

133. The error in a Directional Gyro due to the earth's rotation, at a mean latitude of $45^\circ N$, will cause the spin axis to move by:

- a. 10.6° Clockwise
- b. 10.60 Anti-clockwise
- c. 7.60 Clockwise
- d. 7.60 Anti-clockwise

15 sing
NED

$$15 \times \sin 45 = 10.6$$

134. What are the components of a Ring Laser Gyro?

- a. mirrors and 2 cavities
- b. 2 anodes and 2 cathodes
- c. 2 beams of laser light
- d. horizontal gyro axis and 1 degree of freedom

135. Where on the earth's surface is the earth rate drift of a DGI equal to 15.040 per hour?

- a. 15°
- b. 30°
- c. 0°
- d. 90°

70 KNOTS AVIATION

136. If you maintain the same CAS and Altitude (FL270), and the temperature increases, what happens to the Mach No?

- a. increases at an exponential rate
- b. decreases at an exponential rate
- c. remains the same
- d. increases

doesn't affect Mach number.

137. If CAS is kept constant, what happens to the Mach No?

- a. as the altitude increases the mach No will increase
- b. as the altitude increases the mach No will decrease
- c. as the temperature increases the mach No will increase
- d. as the temperature decreases the mach No will decrease

138. The pendulous type correction detector fitted to the DGI provides:

- a. torque on the sensitive axis
- b. two torque motors on the horizontal axis
- c. pendulous internal nozzle on the outer gimbal
- d. one torque motor

139. An aircraft is fitted with two altimeters. One is corrected for position error, the other is not corrected for position error.

- a. ATC will receive erroneous information of flight level
- b. at high speed the non-compensated altimeter will show a lower altitude
- c. provided that the ADC is working normally, there will be no error to either altimeter
- d. at high speed the non-compensated altimeter will show a higher altitude

140. Density altitude is defined as:

- a. the altitude of the airfield elevation corrected for Lapse Rate
- b. the altitude reading on the altimeter which has QNH set on it
- c. the altitude corresponding to the standard atmosphere compensated for ambient density
- d. the altitude showing on the altimeter with the lowest regional QNH set

141. The pitot tube of an ASI gives a direct reading of:

- a. static pressure
- b. total & static pressure
- c. total pressure
- d. dynamic pressure

142. When descending from FL230 to FL50 at maximum speed, the limitations which apply are:

- a. VMO
- b. VMO then MMO
- c. MMO then VMO
- d. MMO

143. The pressure measured at the forward face of the Pitot probe is:

- a. dynamic pressure
- b. static pressure
- c. total pressure
- d. total pressure + static pressure

144. What has inputs from the flux valve;

- a. error detector
- b. heading indicator
- c. amplifier
- d. precession motor

145. Machmeter readings are subject to which of the following errors;

- a. density error
- b. setting error
- c. temperature error
- d. position/pressure error

146. Sound propagates at a speed which depends only on:

- a. density
- b. temperature
- c. temperature & pressure
- d. pressure

147. What aircraft system uses a frequency of 4,400 MHz?

- a. SSR
- b. radio altimeter
- c. weather radar
- d. ATC radar

148. A low altitude Radio Altimeter, used in precision approaches, has the following characteristics:

- 1. 1540MHz to 1660 MHz range
 - 2. pulse transmissions
 - 3. frequency modulation
 - 4. height range between 0 and 5,000ft
 - 5. an accuracy of +/- 2ft between 0 and 500ft
- a. 1, 4 and 5
 - b. 3 and 4
 - c. 3 and 5
 - d. 2, 3 and 5

149. A modern low altitude Radio Altimeter uses the principle of:

- a. pulse modulated waves, with the difference between the transmitted and received waves displayed on a circular screen.
- b. Frequency modulated waves, where the difference between the transmitted wave and the received wave is measured.
- c. Wave modulation, with frequency shift due to Doppler effect of the ground reflected wave being measured
- d. Triangular wave, with the frequency shift of the ground reflected wave being measured

150. The frequencies used in a low altitude Radio Altimeter are:

- a. 5 GHz to 6 GHz
- b. 5400 MHz and 9400 MHz

- c. 4200 MHz to 4400 MHz
- d. 2700 MHz to 2900 MHz

151. The difference between Magnetic North and True North can be derived by:

- a. deviation curve
- b. deviation card
- c. map with isoclinic lines
- d. map with isogonal lines

152. A direction gyro gets its directional information from:

- a. air data computer
- b. direct reading magnetic compass
- c. flight director
- d. flux valve

153. What is the principle of operation of a VSI:

- a. differential pressure across a capsule
- b. total pressure in a capsule
- c. static pressure in a capsule
- d. dynamic pressure in a capsule

154. In a Remote Indicating Compass, what component feeds the Amplifier?

- a. gyro precession signal
- b. flux valve
- c. annunciator
- d. error detector

155. An aircraft turns right, through 90°, onto North, at 48N, using a direct indicating compass. The aircraft is turning at rate 2. What heading should the aircraft roll out on?

- a. 010°
 - b. 030°
 - c. 330°
 - d. 350°
- Rate 2 - 1 min 6/2 sec*

156. What is the normal operating range of a low altitude Radio Altimeter?

- a. 0 to 2,500ft
- b. 50ft to 2500ft
- c. 0 to 10,000ft
- d. 0 to 7,500ft

157. What is a radio altimeter used for?

- a. to determine aircraft height above mean sea level
- b. to determine aircraft height above ground level

- c. to determine pressure altitude
- d. to determine aircraft altitude

158. Why must Latitude and Longitude be inserted into an IRS?

- a. to determine the aircraft position relative to the earth
- b. to check the IRS position with the Flight Management System
- c. to enable the levelling procedure to commence
- d. to determine the accuracy of the alignment

159. An aircraft is flying a true track of 360° from 50° south to 50° north. What is the average apparent wander rate:

- a. 0° per hour
- b. + 5° per hour
- c. - 5° per hour
- d. depends upon groundspeed

Eg.?

160. You commence a rate 2 turn from south-east to south-west, in the Northern Hemisphere. On what heading do you stop the turn?

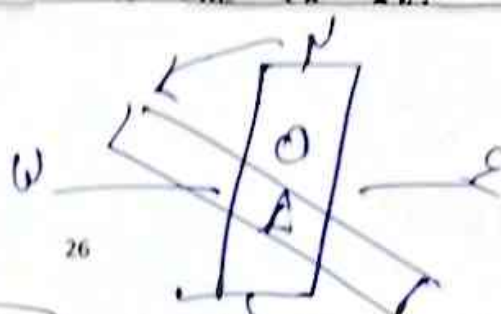
- a. 240°
- b. 255°
- c. 235°
- d. 205°

161. A directional gyro is valid only for a short period of time. The causes of this inaccuracy are:

- 1. earth rotation
- 2. longitudinal accelerations
- 3. a/c motion over the earth
- 4. mechanical defects
- 5. gyro mass
- a. 1, 3 & 5
- b. 1, 3 & 4
- c. 1, 2 & 3
- d. all of the above

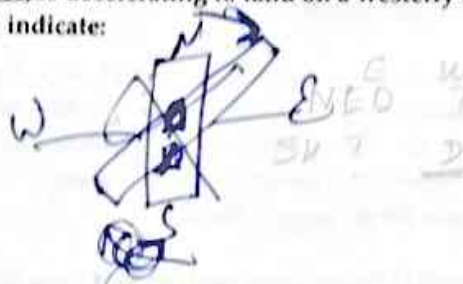
162. A VMO / MMO alarm system, on an airline aircraft, is fitted with an aneroid capsule which is:

- a. subjected to static pressure and an anemometer subjected to dynamic pressure
- b. subjected to dynamic pressure and an anemometer subjected to static pressure
- c. subjected to static pressure and an anemometer subjected to static pressure
- d. subjected to dynamic pressure and an anemometer subjected to dynamic pressure



163. An aircraft, in the southern hemisphere, is decelerating to land on a westerly heading. The direct reading magnetic compass will indicate:

- a. an apparent turn to north
- b. an apparent turn to south
- c. correctly
- d. an oscillation about west



164. What is the input to a VSI?

- a. static pressure
- b. differential pressure
- c. total pressure
- d. dynamic pressure

165. The component(s) used to align an inertial strap-down unit in the horizontal plane is/are:

- a. Accelerometers and gyroscopes
- b. Accelerometers
- c. Flow inductors
- d. Gyroscopes

166. A ring laser gyro consists of:

- a. A gyro with 2 degrees of freedom
- b. Two moving cavities using mirrors
- c. A laser split into two beams
- d. Two electrodes (anodes and cathodes)

167. The Directional Gyro Indicator (DGI) can:

- a. not align itself with magnetic north
- b. can automatically align itself with magnetic north
- c. have 1° of freedom
- d. have 2° of freedom

168. The Pitot tube comprises a mast to position it below the skin of the aircraft for:

- a. avoid disturbance from aerodynamic flow about the aircraft
- b. position it outside the boundary layer
- c. anti-ice protection
- d. easy access for maintenance

169. Using a classic Artificial Horizon, the a/c performs a right turn through 360 degrees at a constant angle of bank and rate of turn. The indication is:

- a. Nose up, too much bank.
- b. Nose up, not enough bank.
- c. Nose up, wings level.
- d. Bank and pitch correct.